Shenghan Gao

1st year M.S.E. supervised by Prof. Quan Li, School of Information Science and Technology, ShanghaiTech University

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Education

ShanghaiTech University — Shanghai, China

M.S.E. in Computer Science and Technology — Sep. 2026 - Now

- Research Interests: **Human-Centered Computing**

ShanghaiTech University — Shanghai, China

B.Eng. in Computer Science and Technology — Sep. 2021 - Jun. 2025

- Research Interests: **Human-Centered Computing**
- GPA: 3.65 / 4.00, Major GPA: 3.95 / 4.00, Graduate Examination Score: 396 / 500.

University of California, Berkeley — Berkeley, CA

Exchange Student in EECS — Aug. 2023 - Dec. 2023

- GPA: 4.00 / 4.00

Research Experience

ViSeer LAB, SIST, ShanghaiTech University — Shanghai, China

Research Intern supervised by *Prof. Quan Li* | Human-Computer Interaction, Data Visualization — Dec. 2022 - Present

My research focuses on bridging the gap between **complex technological systems** and **human cognition** through a dual-lens approach. From a **human factors** perspective, I leverage mixed methods, including qualitative interviews and quantitative behavioral analysis, to **decode user mental models**. Simultaneously, from a **technical** standpoint, I design and develop interactive systems to **optimize cognitive load** and **augment human decision-making**. My current work aims to pioneer new paradigms in **Human-AI Collaboration (HAI)**, with the ultimate goal of driving technological evolution that enhances **human well-being**.

Selected Projects

SCSimulator: An Exploratory Visual Analytics Framework for Partner Selection in Supply Chains through LLM-driven Multi-Agent Simulation

Shenghan Gao, Junye Wang*, Junjie Xiong*, Yun Jiang, Yun Fang, Qifan Hu, Baolong Liu, Quan Li | Accepted by IUI'26 — Jan. 2025 - Feb. 2026

- Proposed a framework, *SCSimulator*, integrating LLM-driven MAS with human-in-the-loop collaboration for partner selection within supply chains.
- Designed a visual analytics system enabling dynamic network exploration, agent-level intervention, and simulation path branching.
- Integrated predictive modeling with SHAP-based XAI and Chain-of-Thought reasoning to provide multi-faceted interpretability of agent decisions and network impacts.
- Validated system effectiveness through quantitative simulation evaluation, a usage scenario and an expert-based user studies.

CommSense: Facilitating Bias-Aware and Reflective Navigation of Online Comments for Rational Judgment

Yang Ouyang*, **Shenghan Gao***, Ruichuan Wang, Hailiang Zhu, Yuheng Shao, Xiaoyu Gu, Quan Li | Accepted by CHI'26 — Jun. 2025 - Jan. 2026

- Investigated how users form judgments in complex comment environments, identifying a four-stage decision path—Initial Framing, Evidence Foraging, Belief Updating, and Synthesis and Judgment—with goal alienation, asymmetric evidence–belief loops, and insufficient synthesis leading to biased outcomes.
- Led co-design workshop to derive four design requirements for bias mitigation in comment navigation interfaces.
- Designed and implemented **CommSense**, an interface-level intervention integrating overview visualization, structured annotation, reflective prompts, and synthesis scaffolding.
- Conducted a user studies demonstrating how *CommSense* improves bias awareness, supports reflective reasoning, and foster evidence-based judgment while maintaining high usability.

From Requirement to Solution: Unveiling Problem-Driven Design Patterns in Visual Analytics

Yuchen Wu, **Shenghan Gao**, Shizhen Zhang, Xiaofeng Dou, Xingbo Wang, Quan Li | Accepted by TVCG — Dec. 2023 - Apr. 2024

- Presented a methodology of meta-analysis for VA research from a problem-driven perspective.
- Contributed a solution typology and refined typologies of requirement and data, formulating updated abstraction frameworks for VA.
- Unveiled problem-solving practice of VA research through a dense, directed, and weighted graph.

Publication

∗ means equal contribution.

[1] **Shenghan Gao**, Junye Wang*, Junjie Xiong*, Yun Jiang, Yun Fang, Qifan Hu, Baolong Liu, Quan Li. SCSimulator: An Exploratory Visual Analytics Framework for Partner Selection in Supply Chains through LLM-driven Multi-Agent Simulation[C]. *Proceedings of the 31st International Conference on Intelligent User Interfaces (IUI '26), 2026.* (**CCF-B**)

[2] Yang Ouyang*, **Shenghan Gao***, Ruichuan Wang, Hailiang Zhu, Yuheng Shao, Xiaoyu Gu, Quan Li. CommSense: Facilitating Bias-Aware and Reflective Navigation of Online Comments for Rational Judgment[C]. *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26), 2026.* (**CCF-A**)

[3] Yuchen Wu, **Shenghan Gao**, Shizhen Zhang, Xiaofeng Dou, Xingbo Wang, Quan Li. From requirement to solution: Unveiling problem-driven design patterns in visual analytics[J]. *IEEE Transactions on Visualization and Computer Graphics (TVCG), 2025.* (**CCF-A**)

[4] Longfei Chen, **Shenghan Gao**, Shiwei Wang, Ken Lin, Yun Wang, Yan Lu, Quan Li. When Systems Take Initiative: A Design Framework for Adaptive, Mixed-initiative Database Querying[C]. *Proceedings of the 2026 ACM Designing Interactive Systems Conference (DIS '26).* (**CCF-C**)

[5] Yang Ouyang, **Shenghan Gao**, Yuheng Shao, Junjie Xiong, Xiyuan Wang, and Quan Li. VidSense: Facilitating Incremental Externalization and Knowledge Consolidation via Canvas-Based Interactions in Video Learning[C]. *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26), 2026.* (**CCF-A**)

[6] Longfei Chen, **Shenghan Gao**, Shiwei Wang, Ken Lin, Yun Wang, Quan Li. QueryGenie: Making LLM-Based Database Querying Transparent and Controllable[C]. *Adjunct Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology(UIST Demo '26), 2025.* (**CCF-A**)

[7] Yuchen Wu, Yuansong Xu, **Shenghan Gao**, Xingbo Wang, Wenkai Song, Zhiheng Nie, Xiaomeng Fan, Quan Li. LiveRetro: Visual analytics for strategic retrospect in livestream e-commerce[J]. *IEEE Transactions on Visualization and Computer Graphics(TVCG/VIS '23), 2023.* (**CCF-A**)

[8] Laixin Xie, Ying Zhang, Xiyuan Wang, Shiyi Liu, **Shenghan Gao**, Xingxing Xing, Wei Wan, Haipeng Zhang, Quan Li. Influence Maximization in Temporal Social Networks with a Cold-Start Problem: A Supervised Approach[C]. *Proceedings of the International AAAI Conference on Web and Social Media(ICWSM '25), 2025.* (**CCF-B**)

Honors

Merit Student of ShanghaiTech University in 2023 - 2024	Dec. 2024
The Special Scholarship for the Undergraduate 3+1 Overseas Exchange Program for the 2023-2024 academic year	Jun. 2024
Merit Student of ShanghaiTech University in 2022 - 2023	Dec. 2023
ChinaVis Data Challenge 2023 Third Prize	Jul. 2023
Merit Student of ShanghaiTech University in 2021 - 2022	Nov. 2022

Services

Conference Reviewers

ACM CHI 2026 (Poster Track)	Feb. 2026
VINCI 2026 (Technical Track)	Aug. 2026
ACM CHI 2027	Oct. 2026

Teaching Assistants

ARTS 1422 - Data Visualization	Feb. 2024 - Jun. 2024
GEHA 1242 - Human-Animal Interaction	Jul. 2024
GEHA 1101 - Cultural Interaction between the East and West along the Silk Road & Maritime Silk Road	Sep. 2024 - Jan. 2025
AI4SCA - AI for Faculty (SCA)	Sep. 2025 - Jan. 2026
GEHA 1003 - Introduction to World Civilizations	Sep. 2024 - Now

Extracurricular Activities

Volunteer Work for ShanghaiTech Welcome Party	Sep. 2021
Student Mentor for Shangdao College	Sep. 2021 - Jun. 2025
Postgraduate Exam Preparation Support Service	Sep. 23, 2026

Skills

Program Language	Javascript, Python, C, C++, Ruby, SQL, etc.
Framework	Vue, D3, jQuery, BootStrap, Rails, etc.
3D Modeling	Blender, Inventor, SolidWorks, etc.
Concept	Human-Computer Interaction, Data Visualization, Data Mining, Agile Development, etc.
Language	Mandarin (Native), English (TOEFL: 102)